



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

NTU Academy for Professional
and Continuing Education

Prompting Foundations

Working reliably with large language models

Module 5

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Getting reliable work out of AI

Prompt engineering for professionals, in twelve ideas

Access is not the skill

Almost everyone can open an AI chatbot. Far fewer get dependable work out of one.

The gap is ***technique, not the tool.***

It predicts. It does not look things up.

An AI writes one plausible word after another. There is no lookup table, and no inner flag for "I don't actually know this".

*On a factual test, one model answered more questions but was wrong **75%** of the time.*

Source: Kalai et al., 2025

A job title is not intelligence

"Act as a senior lawyer" makes the answer sound expert. It does not make it more correct. A persona sets tone.

Across 162 personas and 2,410 questions, adding a persona **did not reliably improve accuracy**.

Source: Zheng et al., 2024

Give it your material

Most questions at work are about your documents, not general knowledge. Paste the source and tell it to answer only from that.

*Even purpose-built research tools still produced unsupported answers **up to a third of the time**.*

Source: Stanford, 2024 (legal-research tools)

Where you put the ask matters

Models read the start and the end of your input best, and skim the middle. Restate your request at the end of a long paste.

Moving the key passage into the middle of a long input **cut accuracy by over 30%.**

Source: Liu et al., "Lost in the Middle", 2024

Make it think, and make it check

For a real decision, ask for the reasoning, not just the verdict. Have it check its own claims. Let it say "I don't know".

Source: Dhuliawala et al., 2023 · Kalai et al., 2025

Two things only you can control

Guard what you paste. The biggest workplace AI risk is confidential material in a consumer chatbot. Samsung engineers leaked source code within about three weeks.

Stay the thinker. *Used as an answer machine, AI can erode the judgment you are building.*

Source: Samsung, 2023 · MIT "Your Brain on ChatGPT", 2025

From a prompt to a spec

A one-off prompt is retyped and forgotten. A structured, reusable spec is saved, shared, and checked.

```
yaml
```

```
role: a support lead
```

```
task: draft a reply to the customer
```

```
output_format: three short, warm sentences
```

Meet Prompt Spec Studio

Your hands-on companion to this session. No coding required.

Step	What you do
Foundations	The mental models from these slides
Learn	The techniques, one at a time
Build	Fill a form, get a spec
Validate	Catch errors before you trust it
Practice	Real work scenarios

Get the most from it

Try before you reveal. The answer sticks better when you attempt it first.

The practice is interleaved on purpose, so you learn to choose the technique, not just apply a named one.

Save what works into your own library.

From asking a chatbot to engineering reliable specs

prompt-spec-studio.vercel.app

Access code: **DSAI-3F-Open**

Version Management

Month/Year	Version	Changes
May 2026	0.0	New Module Design
July 2026	1.0	Updated the deck based on latest research and added concept of Prompt Spec using YAML